



Neural Chaotic Dynamics for Adaptive Exploration Control of an Autonomous Flying Robot

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Abstract. Efficient search and exploration behaviors observed in animals are significant for their survival, encompassing foraging, mate-finding, and predator avoidance. Animal movement patterns underlying the search and exploration behaviors are typically modeled as Lévy flights. Despite their efficacy, current Lévy flight models rely on mathematical formulations and lack connection to biological neural systems. To address this, we propose an adaptive neural system which encodes Lévy distributions for generating and controlling efficient search and exploration behaviors. The neural system consists of three subnetworks: a neural chaotic network, a directional control network, and a step length control network. The neural chaotic network generates chaotic output signals, which are used as exploration seeds, further processed by the directional and step length control networks to control the search and exploration of a flying robot. By simply adjusting the scaling input of the step length control network, our neural control system can generate adaptive exploration behaviors with a combination of global and local search strategies. Our system is implemented on a simulated autonomous flying robot, evaluated across diverse environments, and compared with Gaussian random walk and traditional Lévy flight strategies.

Keywords: Exploration behavior · Navigation control · Chaotic dynamics · Neural dynamics · Neural control · Lévy flight

1 Introduction

Animals' efficient search and exploration in nature are crucial behaviors for their survival, used for activities such as food foraging, finding mates, and avoiding predators. Biological studies have analyzed movement trajectories of various animals, including albatrosses flying over the ocean [2], sharks and other marine predators swimming underwater [10], and *Drosophila* larvae crawling in experimental arenas [9] while performing foraging tasks. These studies indicate that the most common and effective method for such tasks is a Lévy flight (Lévy

walk) model [12]. The model effectively represents the complex behaviors that arise from a combination of local search, which involves collecting nearby food, and global search, which entails moving across large areas to explore new food resources. Inspired by this, several robotic studies have been made to understand and replicate efficient search and exploration behaviors by integrating the Lévy flight model into robotic control systems for environmental exploration [4, 13]. These efforts aim to reproduce exploration behavior based on Lévy flight patterns. However, current Lévy flight models, which are primarily based on mathematical formulas with Lévy distributions, are still difficult to relate to the biological neural systems of animals.

From this point of view, we propose an adaptive neural exploration control system (see Fig. 1(a)) composed of three sub-networks: a neural chaotic network, a directional control network, and a step length control network. The core mechanism relies on the neural chaotic network, a two-neuron recurrent network that generates output signals exhibiting chaotic neural dynamics. These signals are subsequently processed by the directional and step length control networks to finally generate control commands. The directional and step length control commands are applied together to control the exploration behavior of a flying robot. By adjusting the scaling input of the step length control network, we can generate efficient adaptive exploration behaviors exhibiting both global and local search patterns (see Figs. 1(b), 1(c), 1(d)). The proposed neural control system is implemented on an autonomous flying robot (drone) to evaluate its

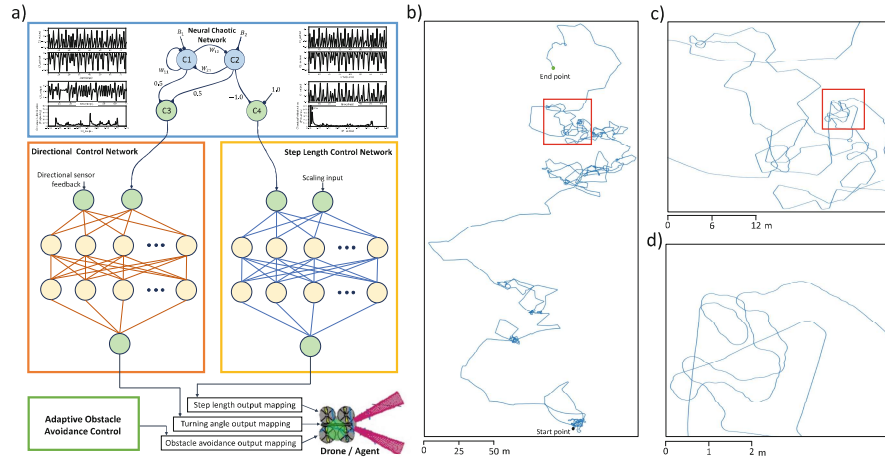


Fig. 1. a) An adaptive neural exploration control system consists of three sub-networks: a neural chaotic network (depicted in blue), a directional control network (depicted in orange), and a step length control network (depicted in yellow). b) A flying trajectory of a flying robot (global search) through an open area. c) Zoomed-in view of the robot's flying trajectory (local search) within the red box in (b). d) Zoomed-in view of the robot's flying trajectory (local search) within the red box in (c). (Color figure online)

performance and demonstrate adaptive exploration behaviors in both open and closed environments of various sizes. These behaviors are compared with two other well-known exploration strategies that use different step length-defined methodologies: Gaussian random walk and Lévy flight.

2 Materials and Methods

This study proposes a method for generating adaptive exploration behaviors using a neural control system, which enables a flying robot to execute both local and global search strategies. The proposed neural control system is based on a neural chaotic network, which exhibits chaotic neural dynamics output. The output is then utilized to produce control commands, which are used for generating adaptive exploration behaviors. The neural chaotic network acts as an internal pattern generator, with its output signals serving two primary functions: directional control via a post-processing directional control network and step length control through a post-processing step length control network. By integrating the outputs from both networks, the flying robot can effectively perform adaptive exploration in various environments.

2.1 Neural Chaotic Network

The neural chaotic network is developed based on a two-neuron recurrent network, as shown in Fig. 2. The neurons $C1$, $C2$, $C3$, and $C4$ in the network are modeled as discrete-time non-spiking neurons where their output is governed by the following equation:

$$o_i(t) = f \left(\sum_{j=1}^n W_{ij} \cdot o_j(t-1) + B_i \right) \quad i = 1, 2, 3, 4, \quad (1)$$

where n denotes the number of neurons, and B_i is the internal bias term to neuron i . W_{ij} represents the synaptic weight of the connection from neuron j to neuron i . $f()$ denotes the activation function, which uses the hyperbolic tangent ($\tanh$) function (for the blue neurons in Fig. 2) and linear function (for the green neurons in Fig. 2).

The network parameters, including synaptic weights and bias inputs, are set according to the neural chaotic system detailed in [3, 11]. Specifically, the synaptic weights are configured as follows: W_{11} is set to -5.5, W_{12} to -1.65, and W_{21} to 1.48. Bias inputs are determined as 5.73 for $B1$ and 0.25 for $B2$. However, initial values of $C1$ and $C2$ are independently generated using random processes. The outputs of $C1$ and $C2$ exhibit chaotic neural dynamics, as shown in Figs. 2(b) and 2(c), characterized by irregularity, aperiodicity, determinism, and high sensitivity to initial conditions [6]. Notably, these outputs are then processed by two linear neurons in distinct ways to produce the directional ($C3$) and step length ($C4$) control inputs (see Figs. 2(d) and 2(g)). $C3$ receives inputs from both $C1$ and $C2$ through synaptic weights of 0.5, while $C4$ solely receives

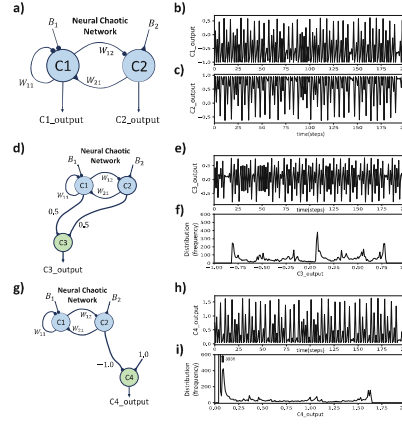


Fig. 2. a) Neural chaotic network. b) Output signal of $C1$. c) Output signal of $C2$. d) Chaotic processing neuron ($C3$) transmitting its output to the directional control network. e) $C3$ output serving as a directional control input. f) Distribution of the directional control input. g) Chaotic processing neuron ($C4$) transmitting its output to the step length control network. h) $C4$ output serving as a step length control input. i) Distribution of the step length control input.

an input from $C2$ through synaptic weight of -1.0 . Additionally, $C4$ also receives a bias input of 1.0 . The outputs of $C3$ and $C4$ exhibit different dynamics, as illustrated in the output signal graphs in Figs. 2(e) and 2(h), and in the output signal distribution graphs in Figs. 2(f) and 2(i), respectively. These output signals ($C3$ and $C4$ outputs) serve as directional and step length control inputs for the directional and step length control networks, described in the following section.

2.2 Directional and Step Length Control Networks

The directional and step length control networks utilized here are constructed based on multilayer feedforward neural networks. All neurons in the networks are modeled as discrete-time non-spiking neurons, as described in Eq. 1, using the linear (for green neurons, Figs. 3(a) and 4(a)) and ReLU (for yellow neurons, Figs. 3(a) and 4(a)) activation functions.

The directional control network, is employed to compute the new heading direction of the drone. This network receives two inputs: the directional control input generated by $C3$ and the current heading directional feedback. The network consists of two hidden layers, each containing 30 neurons, and has a single output representing the drone's new/target heading directional control command.

The synaptic weights in the network are trained using the supervised back-propagation method, with mean absolute error (MAE) as the loss function; see Eqs. 2 and 3. The training dataset is prepared using a state machine for directional computation as described in Eqs. 4 and 5. The training framework and

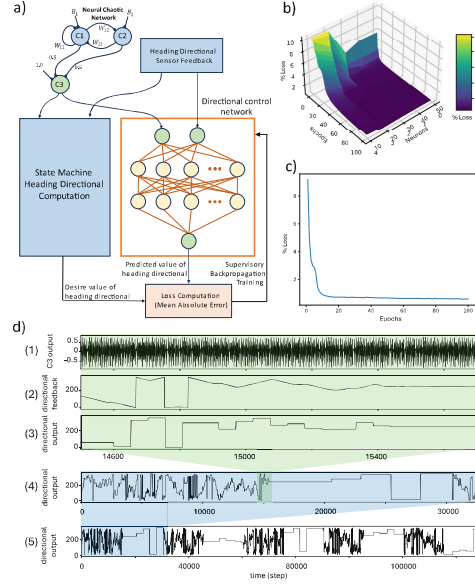


Fig. 3. a) Framework for training the directional control network. b) Surface graph (heatmap) depicting the landscape of loss percentage over the training period. c) Graph showing the loss percentage of the directional control network with 30 neurons for each hidden layer. d) Directional control network input and output signals. The green areas in 1–3 show zoomed-in views of $C3$ output, directional (input) feedback, and directional output from the green area in 4. The blue area in 4 shows a zoomed-in view of the directional output from the blue area in 5, which shows the overall directional output. (Color figure online)

output evaluation are illustrated in Fig. 3.

$$\Delta W_{ij} = -\mu \frac{\partial E}{\partial W_{ij}}, \quad (2)$$

$$E = \frac{1}{n} \sum_{i=1}^n |D_{target-i} - D_{predict-i}|, \quad (3)$$

where W_{ij} denotes the weight of the connection between neuron i and neuron j . μ is the learning rate. E is an error or loss calculated using the mean absolute error. $D_{target-i}$ represents the target heading direction for sample i . $D_{predict-i}$ represents the predicted heading direction for sample i .

$$D_{t-cal}(t) = 90 \cdot o_3(t) + D_c(t), \quad (4)$$

$$D_{target}(t) = \begin{cases} D_{t-cal}(t) + 360 & D_{t-cal}(t) < 0, \\ D_{t-cal}(t) & 0 \leq D_{t-cal}(t) \leq 360, \\ D_{t-cal}(t) - 360 & D_{t-cal}(t) > 360, \end{cases} \quad (5)$$

where D_{t_cal} denotes the calculated target heading direction, o_3 represents the output of $C3$, D_c is the current heading direction value (heading directional feedback), and D_{target} is the final target heading direction.

The step length control network is responsible for computing the step length control command. Empirically, the synaptic weights in this network are set to 2.0. Similar to the directional control network, the step length control network receives two inputs: the step length control input from $C4$ (I_C) and a scaling input (I_S). It consists of two hidden layers, each containing 10 neurons, and has a single output representing the drone's step length control command. The primary function of this network is to scale the step length control input based on the scaling input, which is preprogrammed to cyclically alternate between the empirical values of -1.5 and 3.0. The network architecture and parameter settings allow for simply adjusting the step length according to the scaling input value, enhancing the drone's exploration capabilities. The output (scaling capability) of the network is shown in Fig. 4.

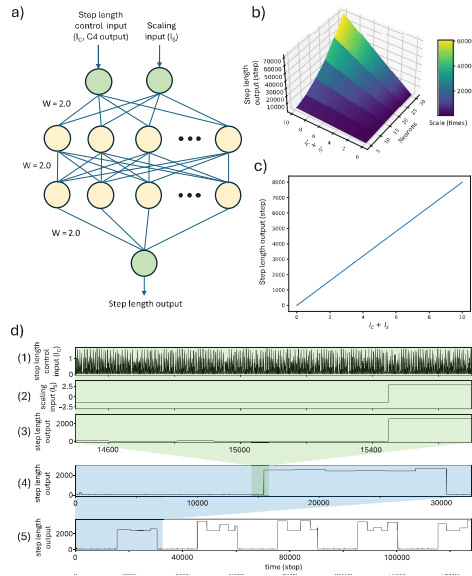


Fig. 4. a) Step length control network. b) Surface graph (heatmap) depicting the output landscape of the network. Note that $I_C + I_S$ represents the summation value of the network inputs. c) Graph showing the output value of the step length control network using ten neurons for each hidden layer. d) Step length control network input and output signals. The green areas in 1–3 show zoomed-in views of inputs I_C , I_S , and step length output from the green area in 4. The blue area in 4 shows a zoomed-in view of the step length output from the blue area in 5, which shows the overall step length output. Here, the step length is based on a computation time step where 100 steps correspond to approximately 0.5 m. (Color figure online)

2.3 Neural System Integration

To achieve adaptive exploration control, we integrate two control commands: directional and step length control commands. These commands are initially generated by the neural chaotic network (Fig. 1(a), blue neurons) and then undergo neural dynamics handling through the linear neurons (Fig. 1(a), green neurons) to ensure that the output signals are suitable for further processing. Subsequently, two multilayer feedforward neural networks (Fig. 1(a), orange and yellow boxes) are employed for post-processing. The overall neural system integration is shown in algorithm 1.

Algorithm 1. Neural System integration

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1: Initialize all variables
2:  $C1, C2 \sim \mathcal{U}(-1.0, 1.0)$ 
3:  $W_{11} = -5.5, W_{12} = -1.65, W_{21} = 1.48, B_1 = -5.73, B_2 = 0.25$ 
4: Deploy all weights of directional control network as trained values
5: Set all weights of step length control network = 2.0
6: while Terminate is not TRUE do
7:   Fly the drone forward
8:   Get current heading directional feedback           ▷ receive input information
9:   Update  $C1$  and  $C2$                                ▷ generate chaotic output signals
10:  Update  $C3$                                        ▷ compute directional control input
11:  Update  $C4$                                        ▷ compute step length control input
12:  if  $Step\_count > Step\_Length\_Command$  then
13:    Compute step length control network           ▷ get new step length command
14:    Compute directional control network           ▷ get new directional command
15:     $Step\_count = 0$ 
16:  end if
17:  if  $Loop < 15000$  then           ▷ perform cyclically alternate the scaling input
18:    if  $Temp \% 2 = 0$  then           ▷ perform local search
19:       $Scaling\_input = -1.5$            ▷ use low scaling input value
20:    else           ▷ perform global search
21:       $Scaling\_input = 3.0$            ▷ use high scaling input value
22:    end if
23:  else
24:     $Temp++$ 
25:     $Step\_count = 0$ 
26:     $Loop = 0$ 
27:  end if
28:  Map the directional command to the turning direction
29:  Execute turning           ▷ turn until achieve the target heading direction
30:   $Step\_count++$ 
31:   $Loop++$ 
32: end while

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3 Experiments and Results

The proposed adaptive neural exploration control system underwent performance evaluation and exploration behavior demonstration using a simulated drone within CoppeliaSim as the experimental platform. A robot operating system (ROS) was used to interface between the drone and the neural control system. To achieve this, two experiments were implemented under five varying sizes environments (see Fig. 5) in both open and closed areas. During the tests, the drone explored the given environment by maintaining a constant forward motion (pitch) at an approximate speed of 0.1 m/s. Concurrently, the heading direction (yaw) was controlled using the proposed adaptive neural exploration control system to facilitate adaptive exploration behaviors.

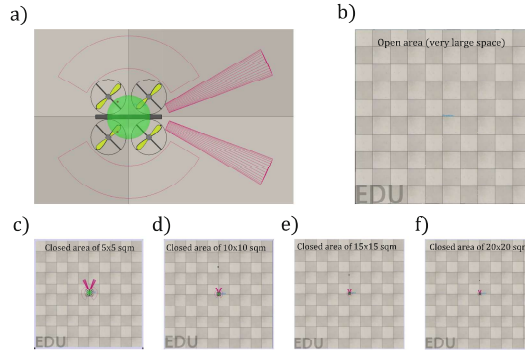


Fig. 5. a) A simulated drone used as the experimental platform. b)-f) Experimental environments with different sizes.

The first experiment aimed to investigate global search performance within an open environment. Three exploration strategies were compared: Gaussian random walk, Lévy flight, and adaptive neural exploration control utilizing neural chaotic dynamics. Each test was approximately 120 min. The results in Fig. 6(a) show three flying trajectories corresponding to the three exploration strategies, all shown within the same map. This visualization allows for the observation of different behaviors compared among each exploration strategy. Specifically, when employing the Gaussian random walk (blue) exploration strategy, the drone performed more looping flight patterns, leading to focused exploration within a specific small area.

In contrast, when utilizing the Lévy flight (orange) and adaptive neural exploration control (green) strategies, the drone exhibited prolonged forward flight patterns. This behavior results in traversing large areas, interspersed with periods of exploration in particular small areas. Figure 6(b) shows step length information compared among each exploration strategy. The results show that the step lengths of the Gaussian random walk strategy consist mostly of low-range values, although with the highest frequency. While for both the Lévy

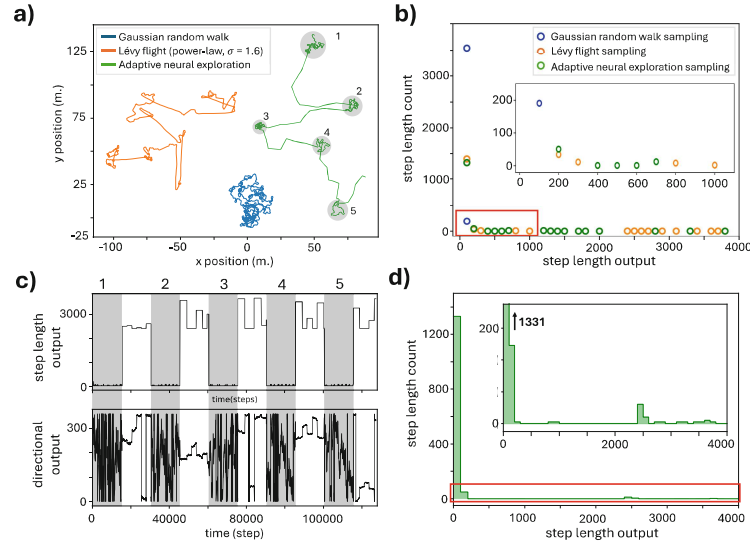


Fig. 6. a) Comparison of flying trajectories among three exploration control strategies: Gaussian random walk (depicted in blue), Lévy flight (depicted in orange), and adaptive neural exploration control (depicted in green). b) Comparison of step length information. The inset plot shows a zoomed-in view of the red box. c) Step length and directional control output signals. The shaded gray areas represent the output signals corresponding to the shaded flying patterns in (a). d) Step length histogram of the adaptive neural exploration control strategy. The inset plot shows a zoomed-in view of the red box. A video showing adaptive exploration flying behaviors can be seen at www.manoonpong.com/AEC/video.mp4. (Color figure online)

flight and adaptive neural exploration control strategies, their step lengths were predominantly low-range values at a moderate frequency, and the rest was distributed across high-range values. Additionally, Fig. 6(c) shows the step length and directional output signals. The shaded gray areas represent the output signals with low step length output values and high-frequency directional outputs, corresponding to the shaded flying trajectory in Fig. 6(a), which exhibits the local search pattern. Figure 6(d) shows the histogram of the step length using the adaptive neural exploration control strategy. Based on the step length distribution and histogram of the adaptive neural exploration control in Figs. 6(b) and 6(d), we conducted a Kolmogorov-Smirnov test to assess the distribution similarity between the adaptive neural exploration control and the Lévy flight (power-law model with exponent $\alpha = 1.6$). The result shows that there was no significant difference between the two distributions at $p > 0.05$.

The second experiment focused on assessing local search performance within four closed environments, with map sizes of 5×5 , 10×10 , 15×15 , and 20×20 sqm. The experiments were conducted for approximately 10, 20, 30, and 40 min, respectively, according to the map sizes. Each map underwent testing from three

initial angles (0, 120, and 240°), and the experiments were repeated ten times for each map. The results in Figs. 7(a), 7(b), and 7(c) provide a comparison of the flying trajectories for each exploration strategy in each map. The gray boxes in the graphs indicate the unexplored grid areas.

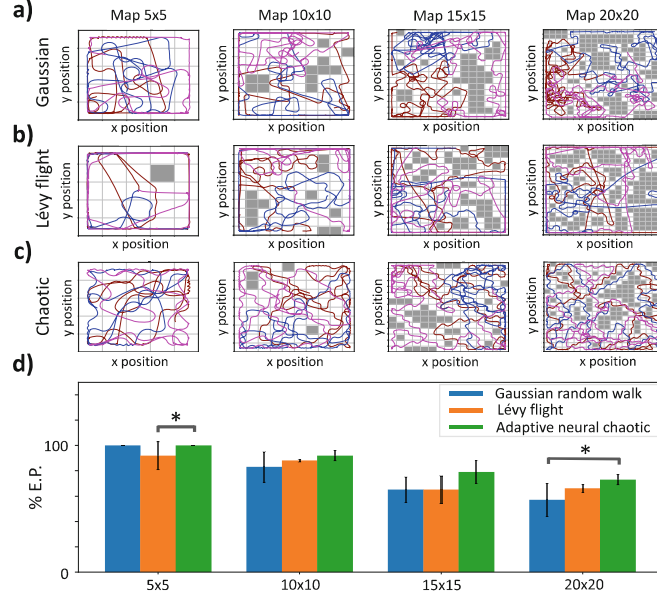


Fig. 7. Drone flying trajectories in closed areas of sizes 5×5 , 10×10 , 15×15 , and 20×20 sqm using three exploration control techniques: a) Gaussian random walk, b) Lévy flight, and c) adaptive neural (chaotic) exploration control. The red, blue, and pink lines represent three initial angles (0, 120, and 240°). d) Comparison of exploration performance percentage (%E.P.) among each map and technique. * denote significant differences with the following $p \leq 0.05$. (Color figure online)

Figure 7(d) illustrates a comparison of exploration performance percentage (%E.P.), calculated as below:

$$\% E.P. = \left(\frac{C_A}{A} \right) \left(\frac{C_G}{A} \right) \times 100\%, \quad (6)$$

where A denotes the total exploration area (grid count). C_A is the largest covered area. C_G is the total covered grid or covered density. Note that grid resolution directly affects the exploration performance percentage. Here, we used a grid size of 1×1 sqm.

The overall result, as depicted in Fig. 7(d), indicates that the proposed adaptive neural exploration control consistently outperformed the other two exploration strategies across all maps. Interestingly, the performance of the Gaussian random walk exploration strategy significantly decreased in a map size of

20×20 sqm (a large closed environment), suggesting its efficiency for exploration in smaller environments. Conversely, the performance of the Lévy flight exploration strategy dropped notably in a map size of 5×5 sqm (a small closed environment), indicating its suitability for larger environment exploration. This observation emphasizes the versatility of our proposed adaptive neural exploration control, which surpasses both Gaussian random walk and Lévy flight strategies in both small and large environments. Hence, it emerges as the most efficient exploration strategy for environments of all sizes.

4 Discussion and Conclusion

In this study, we introduced an adaptive neural exploration control system designed to generate adaptive exploration behaviors for an autonomous drone. The proposed system leverages i) the neural chaotic dynamics of a small recurrent network to generate complex patterns, and ii) nonlinear transformations of multilayer feedforward networks to translate these patterns into appropriate directional and step length control commands for the drone. Additionally, at an abstract level, the proposed system is comparable to a biological neural system in two components: first, an internal pattern generator [1, 5, 9], which here is a neural (chaotic) pattern generator network, and second, premotor control networks [7], which are here realized as directional and step length premotor control networks.

This neural exploration control strategy enables the drone to perform both global search, by traversing large areas, and local search, by intermittently investigating specific regions. These flying behaviors mimic the efficient exploration patterns observed in nature [2, 9, 10], surpassing the performance of both Gaussian random walk and Lévy flight strategies across all closed environments.

Future work will focus on integrating adaptive neural obstacle avoidance control [8] with the exploration control system (see Fig. 1(a), green box). This integration will enable safe exploration missions by allowing the drone to avoid obstacles. We will then validate the combined system on a real drone. Additionally, we will investigate the use of sensor feedback for adaptive step-length control, potentially refining the balance between global and local search strategies.

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Non-instructed Motor Skill Learning in Monkeys: Insights from Deep Reinforcement Learning Models

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Abstract. We employ Reinforcement Learning (RL) models to unravel the mechanisms behind the learning behavior of two macaque monkeys engaged in a free-moving multi-target reaching task. The study was conducted using computer simulations reflecting the animal's learning conditions, and compared with the actual arm movements recorded on two macaque monkeys on a Kinarm apparatus. Our paper thus provides important insights for the design of motor control learning systems, combining end-effector control design with the learning of motor chunks associations. Our research is of interest for the modeling and understanding of natural motor learning systems, but also heads toward the design of more “brain-inspired” adaptive robotic manipulators.

Keywords: Animal Behavior and Cognition · Computational neuroscience · Reinforcement learning · Motor control

1 Introduction

The survival of animals in complex environments relies on their ability to continually learn and refine their skills to enhance both efficiency and competitiveness. [11, 12]. Motor skill learning takes place over long time scales (from weeks to years), reflecting the slow acquisition of combination/coordination rules of the motor apparatus to control its many degrees of freedom [4, 8, 12, 30]. Complex skills generally require a specific combination of elementary motor commands [16, 23, 24, 26], and involves visual/perceptual monitoring at different steps of the process [27, 31]. The shaping of composite actions, made of many elementary movements, has led experimentalists to formulate the *chunking hypothesis*, which